

System Topology & Flow Diagram (English)

— v3.0

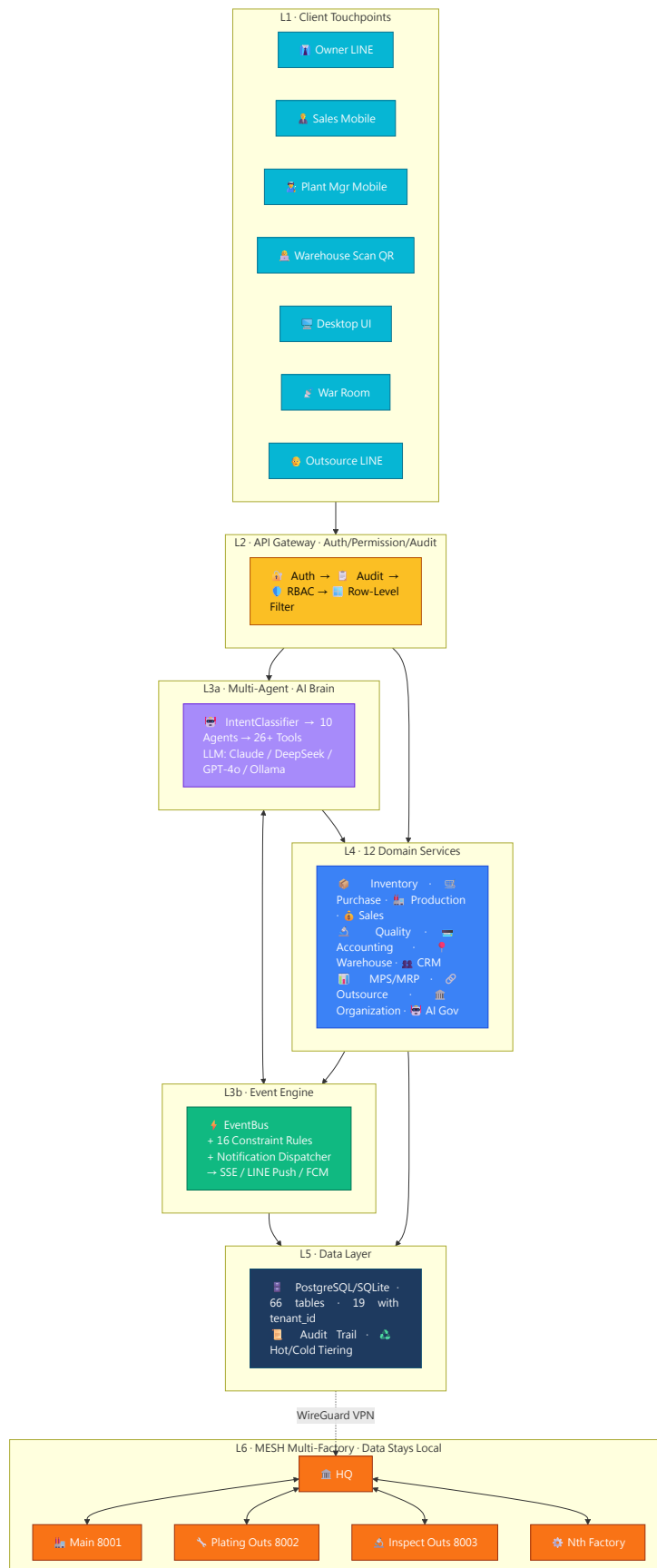
Six perspectives from shallow to deep, static to dynamic — making LLM-ERP understandable to both technical and non-technical readers.

- **Visual:** [system_flow_topology.svg](#) (beautiful SVG for slides / A3 print)
- **Static legacy:** [architecture_diagram.svg](#) (5-layer)

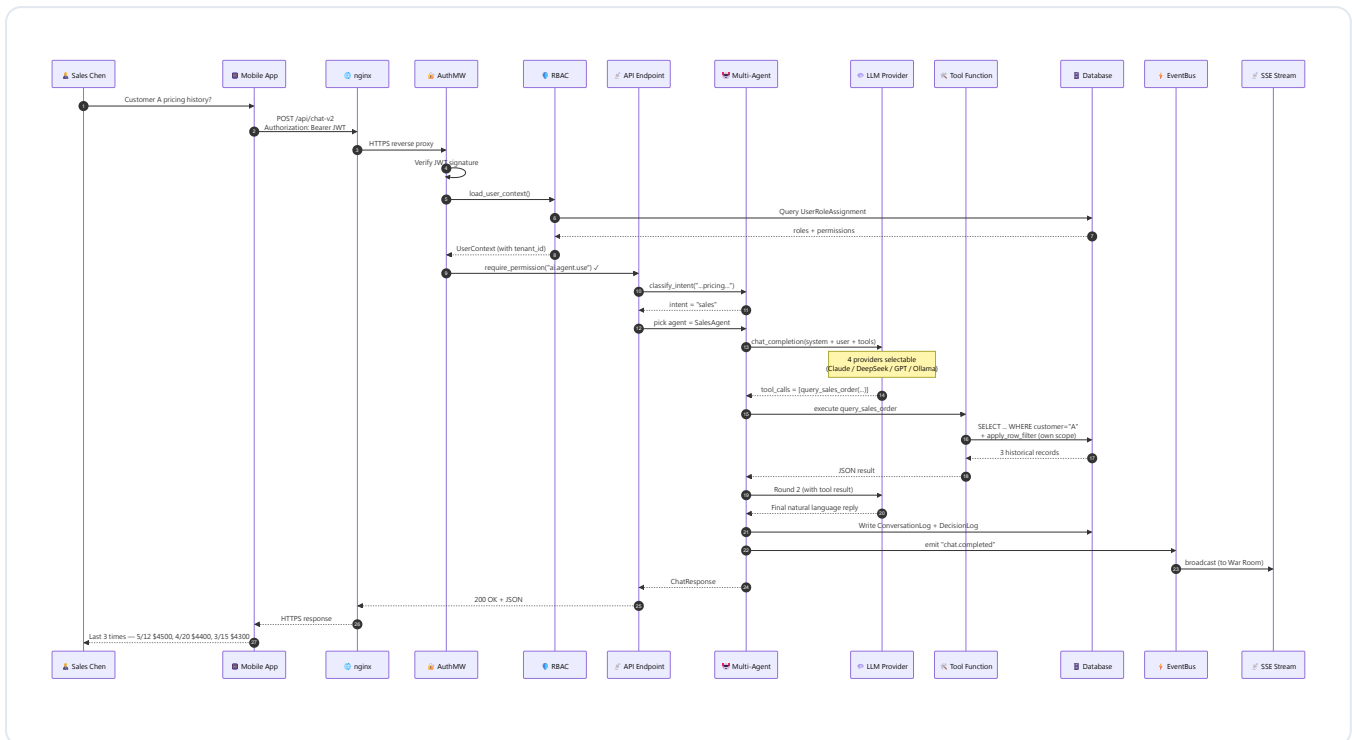
Chinese version: [SYSTEM_TOPOLOGY_ZH.md](#)

⚡ **v3.0 Strategic Pivot Notice:** SVG may still show "🤖 Outsource LINE / Plating Outsource / Inspection Outsource" nodes. v3.0 deprecates outsource / LINE / Mobile (moved to Phase 7). SVG redraw pending Phase 1 completion.

View 1: Six-Layer Overall Architecture (Owner 30-sec scan)



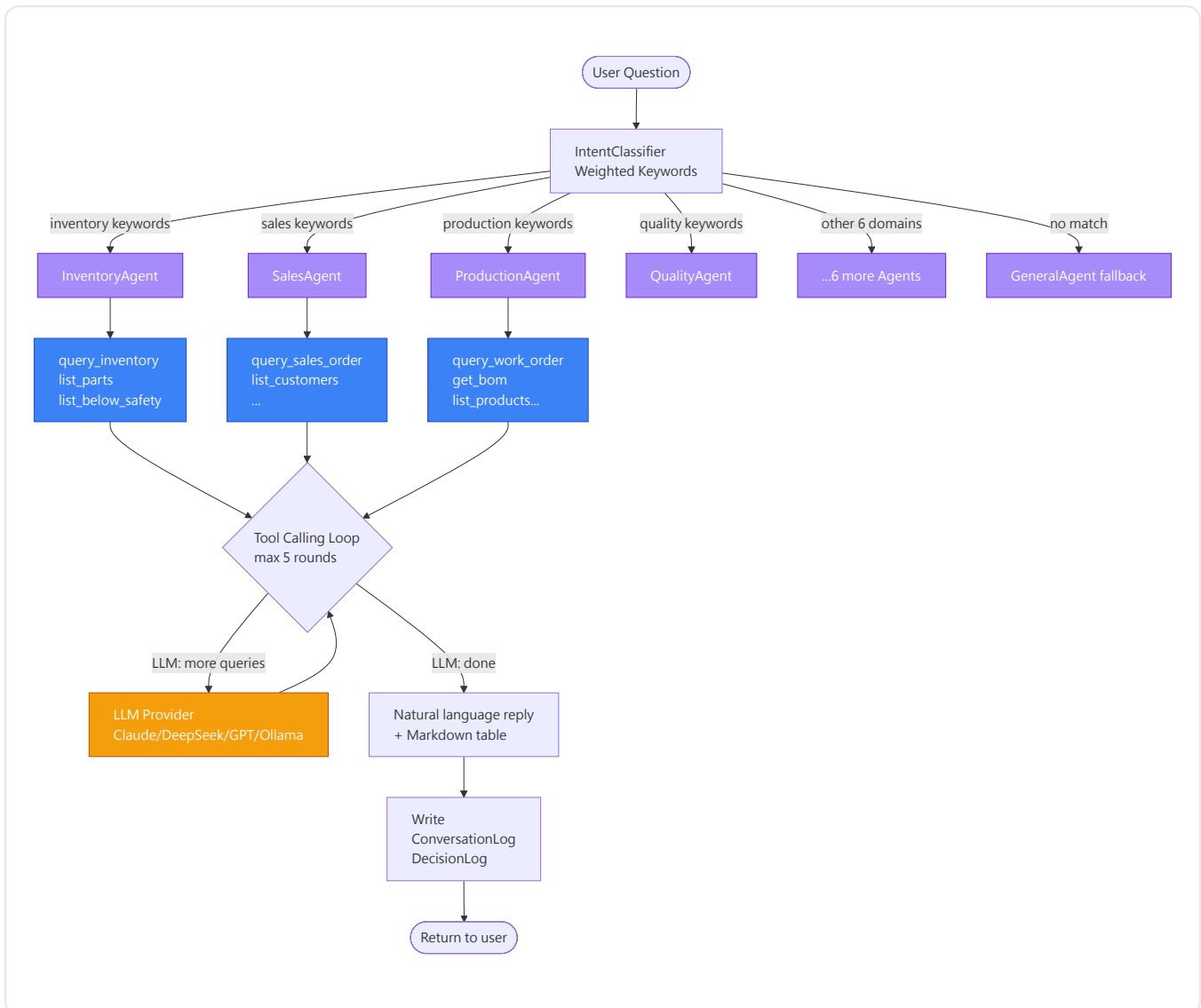
View 2: Full Request Lifecycle (Engineer "how data flows")



Key flow notes:

1. Steps 6-7: Permission load = single JOIN query, 5-min TTL cache
2. Step 11: IntentClassifier uses weighted keywords
3. Step 14: LLM tool calling can loop up to 5 rounds
4. Step 18: `apply_row_filter` auto-adds `WHERE created_by = chen` (Chen sees only own customers)
5. Step 23: All AI decisions written to DecisionLog for audit

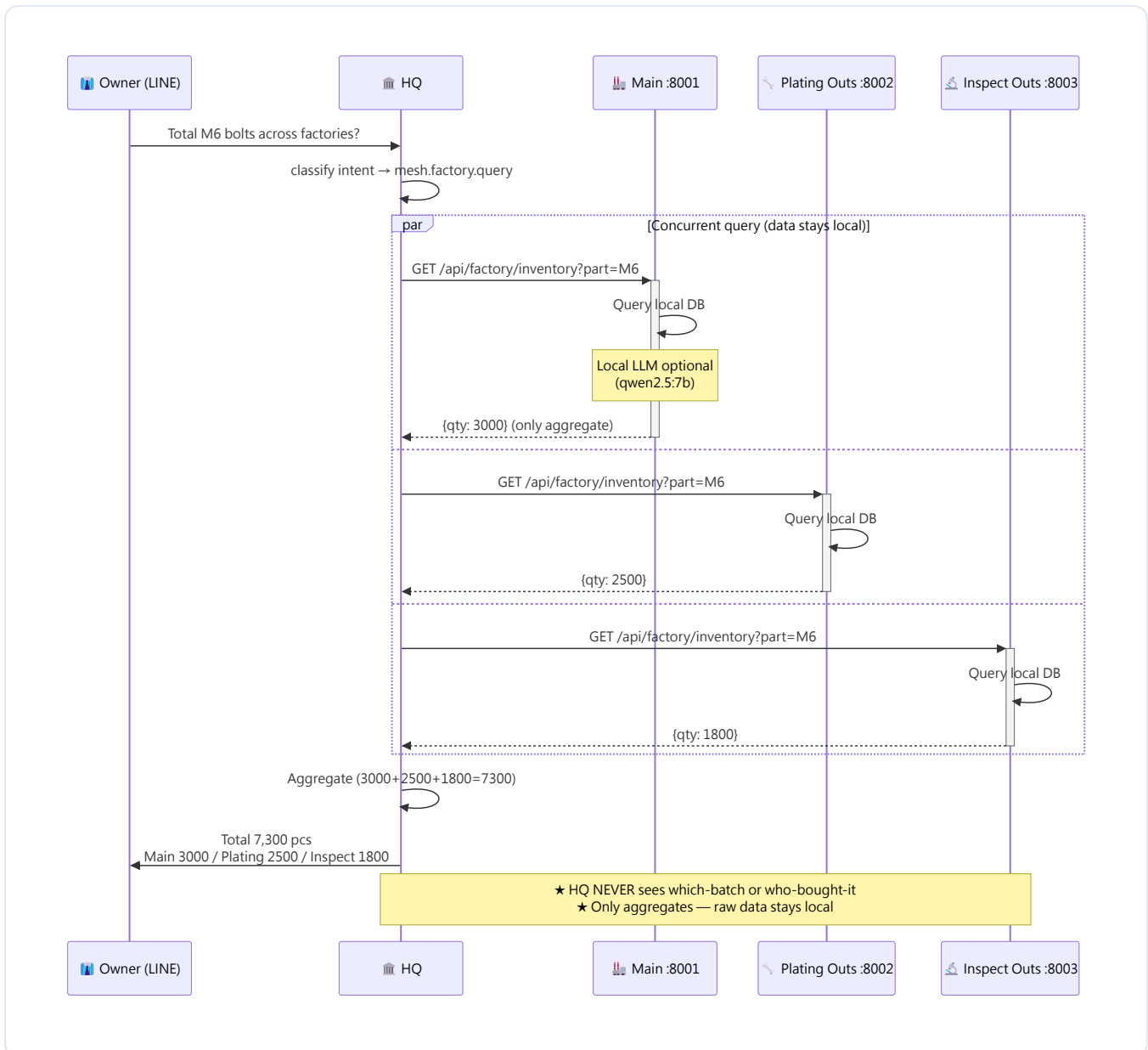
View 3: Multi-Agent Internals (AI Engineer)



Each Agent has scoped tools — preventing LLM from misfiring cross-domain:

- **InventoryAgent** only sees 4 inventory tools
- **SalesAgent** only sees 4 sales tools
- → Inventory agent cannot accidentally trigger sales operations

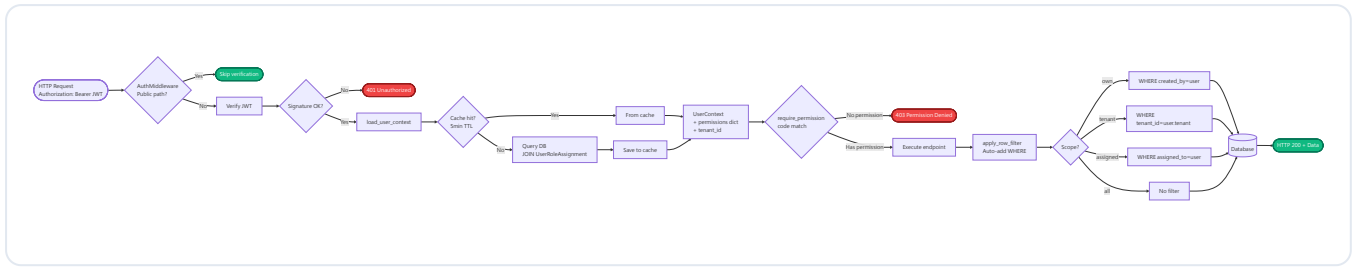
View 4: MESH Multi-Factory Flow (Multi-Site Owner)



Three MESH highlights:

1. **Data sovereignty:** Each factory's data physically stays on-premise
2. **Offline-capable:** Factory keeps working even when HQ disconnects
3. **Infinite scale:** Adding the Nth outsource partner takes 50 lines of config

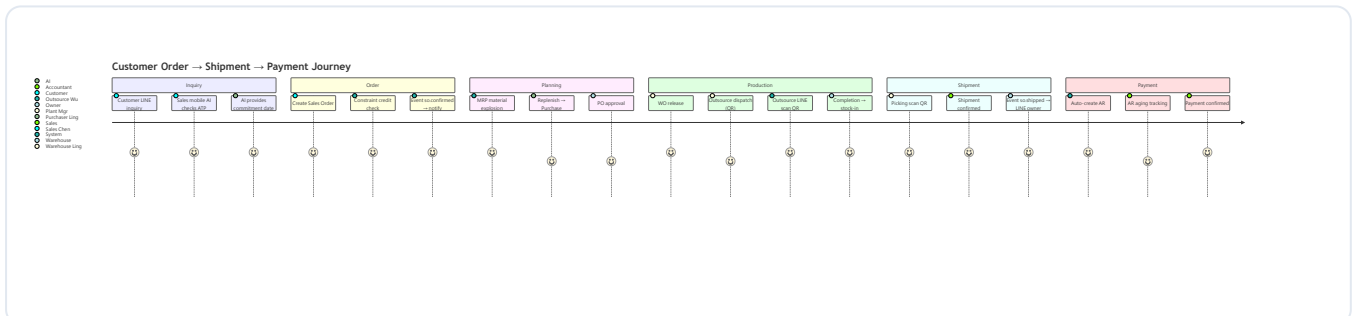
View 5: Permission Check Flow (Security / IT)



Five layers of permission gates:

1. **JWT verification:** Is the token valid
2. **UserContext load:** Who are you, what role
3. **require_permission:** Can you do this action
4. **apply_row_filter:** Which rows can you see (Sales A can't see Sales B's customers)
5. **Audit Trail:** Record who did what when

View 6: Typical Business Lifecycle (Inquiry → Payment)



Events triggered at each stage:

Stage	Event	Auto Action
Inquiry	chat.completed	DecisionLog records AI recommendation
Order	so.confirmed	Notify production_manager
Planning	mrp.generated	Push to purchaser
Purchase	po.approved	Notify supplier + warehouse
Production	wo.released	Push to plant_manager + operators
Outsource	outsource.completed	Push to warehouse + accounting
Shipment	so.shipped	LINE push to owner, auto-create AR
Payment	payment.received	Notify sales + accounting

Tech / Business Dual-Track Summary

Dimension	Tech View	Business View
L1 Client	React/Expo/HTML+SSE	Owner / Sales / Plant Mgr / Outsource
L2 API Gateway	FastAPI Middleware Stack	"Identity verified before entry"
L3a Multi-Agent	IntentClassifier + Tool Calling	"AI auto-finds the right function"
L3b Event Engine	EventBus + Constraint Rules	"Anomalies push automatically"
L4 Domain	12 service modules	"12 business domains integrated"
L5 Data	PostgreSQL + Row-Level	"Data isolated by factory & user"
L6 MESH	WireGuard + Aggregate Query	"Outsource data stays at outsource"

Reading Order by Role

Reader	Suggested Order
 Owner	View 1 (30s scan) → View 6 (business journey)
 Sales	View 6 → View 2 (one request lifecycle)
 Developer	View 2 → View 3 (Multi-Agent) → View 5 (Permission)
 IT/Security	View 5 (Permission) → View 4 (MESH)
 Multi-Site Mgr	View 4 (MESH) → View 1 (overview)

Related Documents

-  [ARCHITECTURE_DIAGRAM.md](#) — Static 5-layer
-  [NETWORK_DEPLOYMENT_EN.md](#) — Network deployment
-  [PERMISSION_MODEL.md](#) — Permission model
-  [LLM_BENCHMARK_REPORT_EN.md](#) — LLM benchmark
-  [ARCHITECTURE_DECISIONS.md](#) — ADRs

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